

CURRICULUM VITAE: JAKE CALLAHAN

PhD Candidate, Program in Applied Mathematics, The University of Arizona

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EDUCATION

Doctor of Philosophy, Applied Mathematics

2023–present

Program in Applied Mathematics, The University of Arizona

Committee: Jason Pacheco (Chair), Tommie Catanach, Chicheng Zhang, Joseph Watkins

Master of Science, Mathematics

2023

Department of Mathematics, Brigham Young University

Hamiltonian Monte Carlo for Reconstructing Historical Earthquake-induced Tsunamis

Committee: Jared Whitehead (Chair), Tyler Jarvis, Mark Kempton

Bachelor of Science, Mathematics (Applied and Computational Emphasis)

2020

Department of Mathematics, Brigham Young University

PUBLICATIONS

CONFERENCE PUBLICATIONS

- [9] J. CALLAHAN, AND T. CATANACH, *On the Misinformation in a Statistical Experiment*, *The 29th International Conference on Artificial Intelligence and Statistics*, 2026, <https://openreview.net/forum?id=k066LSLmB8>.
- [8] J. CALLAHAN, A. CHIN, J. PACHECO, AND T. CATANACH, *Reverse-Annealed Sequential Monte Carlo for Efficient Bayesian Optimal Experiment Design*, *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025, <https://doi.org/10.52202/085713-2176>.

JOURNAL PUBLICATIONS

- [7] J. CALLAHAN, K. MONOGUE, R. VILLAREAL, AND T. CATANACH, *Analysis and Optimization of Seismo-Acoustic Monitoring Networks with Bayesian Optimal Experimental Design*, *Geophysical Journal International* 240 (3) (2025), pp. 1802–1824, <https://doi.org/10.1093/gji/ggae458>.
- [6] A. KING, J. MURRI, J. CALLAHAN, A. RUSSELL, AND T. J. JARVIS, *Mathematical Analysis of Redistricting in Utah*, *Statistics and Public Policy* 9 (1) (2022), pp. 136–148, <https://doi.org/10.1080/2330443X.2022.2105770>.

IN REVIEW

- [5] J. CALLAHAN, T. LUCKIE, R. PORRITT, AND T. CATANACH, *Optimizing Constrained Seismic Monitoring Networks under Model Uncertainty*, Submitted to *Inverse Problems* (2026).

IN PREPARATION

- [4] J. CALLAHAN, AND J. PACHECO, *Distributional Reinforcement Learning for Stable and Flexible Sequential Bayesian Optimal Experimental Design*, 2026.

SOFTWARE

- [3] J. CALLAHAN, K. MONOGUE, R. VILLAREAL, AND T. CATANACH, *Seismic optimal experimental design in Python*, https://github.com/sandialabs/seismic_boed.

TECHNICAL REPORTS

- [2] J. CALLAHAN, AND T. CATANACH, *Importance Sampling in Bayesian OED for Sensor Placement*, *Computer Science Research Institute Summer Proceedings 2021*, The Computer Science Research Institute at Sandia National Laboratories, Albuquerque, NM, 2021, pp. 283–292, https://bit.ly/csri_2021.

OTHER PUBLICATIONS

- [1] J. CALLAHAN, “Hamiltonian Monte Carlo for Reconstructing Historical Earthquake-Induced Tsunamis”, MA thesis, Brigham Young University, 2023, <https://scholarsarchive.byu.edu/etd/9950/>.

PRESENTATIONS

CONFERENCE PRESENTATIONS

- [10] *Bayesian OED for Sensor Placement: Analysis and Optimization of Seismo-Acoustic Monitoring Networks with Bayesian Optimal Experimental Design*, SIAM Conference on Computer Science and Engineering, Amsterdam, The Netherlands, Mar. 2023.
- [9] *Markov-chain Monte Carlo Methods for Reconstructing Historical Earthquake-induced Tsunamis*, SIAM Northern States Sectional Conference, Logan, UT, USA, Apr. 2023.
- [8] *Markov-chain Monte Carlo Methods for Reconstructing Historical Earthquake-induced Tsunamis*, Brigham Young University Student Research Conference, Provo, UT, USA, Apr. 2023.
- [7] *Markov-chain Monte Carlo Methods for Reconstructing Historical Earthquake-induced Tsunamis*, Geological Society of America Connects, Denver, CO, USA, Oct. 2022, https://bit.ly/gsa_2022.

SEMINARS AND INVITED TALKS

- [6] *Bayesian Optimal Experimental Design for Inferring Kinetic Monte Carlo Parameters in Fusion-Irradiated Materials*, Arizona-Livermore Days, Tucson, AZ, 2026.
- [5] *Wrong by Design: How to Choose Experiments with an Imperfect Model*, Program in Applied Mathematics Brown Bag Seminar, The University of Arizona, Tucson, AZ, 2026.
- [4] *Bayesian Optimal Experiment Design for Analysis and Optimization of Seismo-Acoustic Monitoring Networks*, Program in Applied Mathematics Brown Bag Seminar, The University of Arizona, Tucson, AZ, 2024.

POSTER PRESENTATIONS

- [3] *On the Measure of the Misinformation in a Statistical Experiment*, The 29th International Conference on Artificial Intelligence and Statistics, Tangier, Morocco, 2026.

- [2] *Reverse-Annealed Sequential Monte Carlo for Efficient Bayesian Optimal Experimental Design*, The Thirty-ninth Annual Conference on Advances in Neural Information Processing Systems, San Diego, CA, USA, 2025.
- [1] *Markov-chain Monte Carlo Methods for Reconstructing Historical Earthquake-induced Tsunamis*, American Geological Union Fall Meeting, Chicago, IL, USA, Dec. 2022, https://bit.ly/agu_2022.

AWARDS AND HONORS

Herbert E. Carter Travel Award 2026
 Graduate Interdisciplinary Programs, University of Arizona

TEACHING

INSTRUCTOR

- Math 107: Exploring and Understanding Data University of Arizona, Fall 2025
- Math 112: College Algebra University of Arizona, Fall 2024
- Math 102: Quantitative Reasoning Brigham Young University, Spring 2022

GRADUATE TEACHING ASSISTANT

- Math 466: Theory of Statistics University of Arizona, Fall 2025
- Math 584: Intro to Real Analysis University of Arizona, Fall 2024–Spring 2025
- Math 112: College Algebra University of Arizona, Fall 2023–Spring 2024
- Math 342: Theory of Analysis 2 Brigham Young University, Fall 2021–Spring 2023
- Math 112: Calculus 2 Brigham Young University, Fall 2021

PROFESSIONAL SERVICE

DEPARTMENTAL AND UNIVERSITY SERVICE

Student Ambassador, NSF Innovations in Graduate Education (IGE) 2025 – 2026

- Facilitated partnerships between the University of Arizona Program in Applied Mathematics and Sandia National Laboratories to promote non-traditional research careers in National and Industrial Labs.

Organizer and Instructor, U of A Applied Math Programming Workshop 2024

- Designed and led a week-long workshop to teach 20+ incoming PhD students Python and scientific computing best practices.

PROFESSIONAL AND TECHNICAL SERVICE

Journal Referee 2025–Present

- Seismological Research Letters
- Geophysical Journal International
- Neural Information Processing Systems

Mini-symposium Organizer, SIAM Northern States Sectional 2023

- *Is Quantifiable Uncertainty Achievable? A (mostly) Bayesian Perspective.* Logan, UT.